Test 1 this Thursday

Test 1 (during the tutorial) covers material from Notes 1-4 (chapter 1 of the textbook). It contains four multiple choice questions (similar to online quiz questions) and two questions which require detailed written steps (similar to tutorial questions).

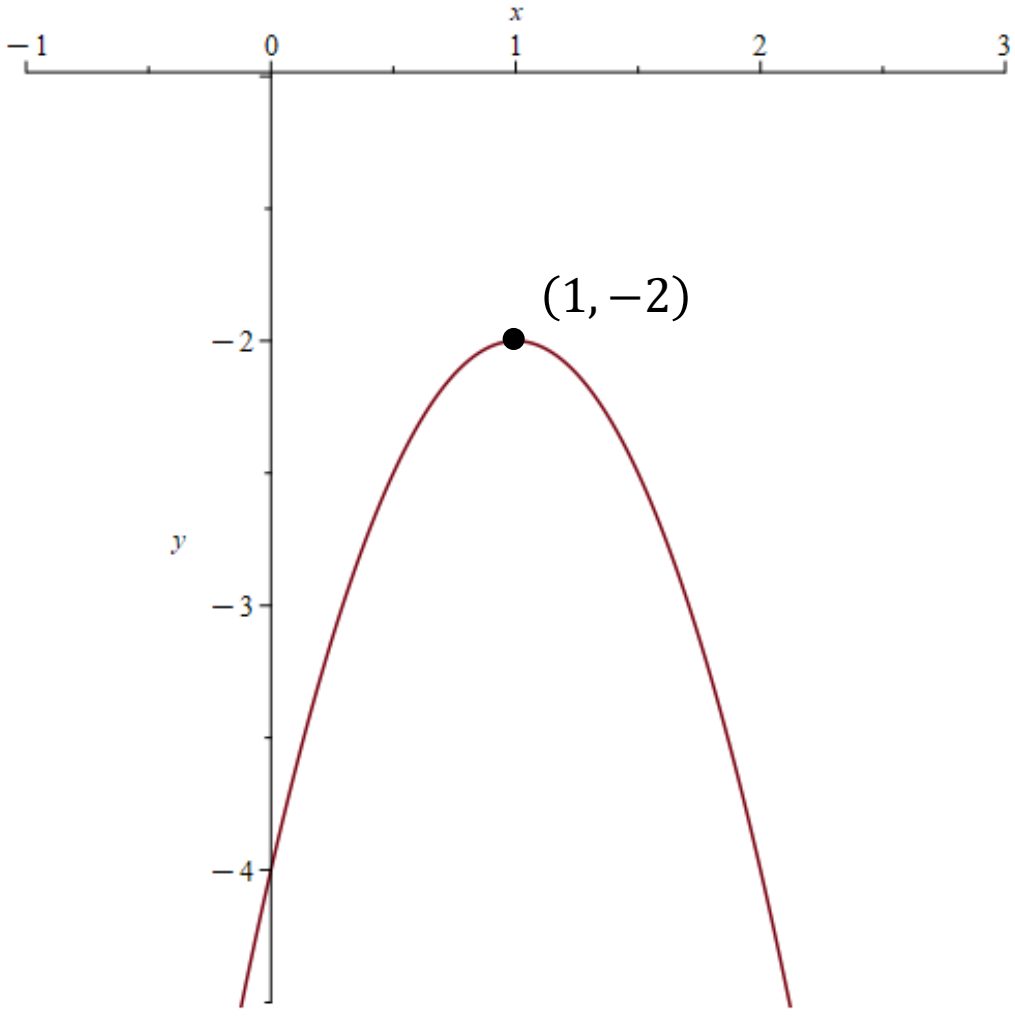
Quadratic Functions

Example Plot the graph of $y = -2x^2 + 4x - 4$. Find its vertex, y-intercept and x-intercepts (if there is any).
Solution We first rewrite the quadratic function in the vertex form (with $a = -2, b = 4, c = -4$):

$$\begin{aligned} -2x^2 + 4x - 4 &= -2\left(x - 1\right)^2 + \frac{4 \times (-2) \times (-4) - 4^2}{4 \times (-2)} \\ &= -2\left(x - 1\right)^2 - 2. \end{aligned}$$

Hence the vertex is $(1, -2)$. The y-intercept is $(0, -4)$. Since the parabola opens downward with vertex $(1, -2)$, it lies entirely below the x-axis. So, there is no x-intercept.

The graph of $y = -2x^2 + 4x - 3$



$$y = -2x^2 + 4x - 4$$

Appendix B.5 Complex Numbers

Suppose we try to solve the quadratic equation

$$-2x^2 + 4x - 4 = 0.$$

That is,

$$-2\left(x-1\right)^2 - 2 = 0.$$

We obtain

$$\left(x-1\right)^2 = -1.$$

Since the square of any real number is always non-negative, the above equation has no real number solution. We may introduce the imaginary number i such that

$$i = \sqrt{-1}, \text{ or } \ i^2 = -1.$$

Appendix B.5 Complex Numbers

In general, we use the vertex form

$$ax^2 + bx + c = a\left(x + \frac{b}{2a}\right)^2 + c - \frac{b^2}{4a}$$

to solve the quadratic equation

$$ax^2 + bx + c = 0. \tag{1}$$

We have

$$\left(x + \frac{b}{2a}\right)^2 = \frac{b^2 - 4ac}{4a^2}.$$

If $b^2 - 4ac > 0$, we obtain two distinct real number solutions

$$-\frac{b}{2a} \pm \frac{\sqrt{b^2 - 4ac}}{2a}.$$

Appendix B.5 Complex Numbers

If $b^2 - 4ac = 0$, we obtain two identical real number solutions

$$-\frac{b}{2a}.$$

If $b^2 - 4ac < 0$, we may write

$$\sqrt{b^2 - 4ac} = \sqrt{(-1)(4ac - b^2)} = \sqrt{4ac - b^2}\ i.$$

The solutions to equation (1) can be written as

$$-\frac{b}{2a} \pm \frac{\sqrt{4ac - b^2}}{2a}\ i,$$

which are *complex numbers*.

Appendix B.5 Complex Numbers

Example Solve the quadratic equation $x^2 + 2x + 5 = 0$.
Solution We have $a = 1, b = 2, c = 5$, and $b^2 - 4ac = -16 < 0$. Hence the solutions are

$$-\frac{2}{2} \pm \frac{\sqrt{16}}{2}\, i = -1 \pm 2i,$$

which are two distinct complex numbers. We may verify the solutions as follows.

$$\begin{aligned}
&(-1 + 2i)^2 + 2(-1 + 2i) + 5 \\
&= (-1)^2 - 4i + (2i)^2 - 2 + 4i + 5 \\
&= 4 + 4i^2 \\
&= 4 + 4(-1) \\
&= 0.
\end{aligned}$$

Appendix B.5 Complex Numbers

We also have

$$\begin{aligned}
&(-1-2i)^2+2(-1-2i)+5\\
&=(-1)^2+4i+(-2i)^2-2-4i+5\\
&=4+4i^2\\
&=0.
\end{aligned}$$

In general, a complex number z in standard form is written as $z=a+bi$, where a and b are real numbers.
The number a is the *real part* of z, denoted by $\mathrm{Re}\ z$, and b is the *imaginary part* of z, denoted by $\mathrm{Im}\ z$.
The complex number $a-bi$ is called the *conjugate* of $z=a+bi$, and is denoted by $\bar{z}$.
Thus, the solutions to a quadratic equation form conjugate pairs.

Appendix B.5 Complex Numbers

Two complex numbers are equal if and only if their real parts are equal, and their imaginary parts are equal. Arithmetic operations of complex numbers follow the same rules as for real numbers:

$$\begin{aligned}(a+bi)+(c+di) &= (a+c)+(b+d)i\\ (a+bi)-(c+di) &= (a-c)+(b-d)i,\\ (a+bi)(c+di) &= ac+adi+bci+bdi^2\\ &= (ac-bd)+(ad+bc)i.\end{aligned}$$

In particular, we have

$$(a+bi)(a-bi) = a^2+b^2.$$

Appendix B.5 Complex numbers

For a complex number in standard form $z = a + bi$, We have

$$z + \bar{z} = 2a,\ \ z - \bar{z} = 2bi,\ \ z\bar{z} = a^2 + b^2.$$

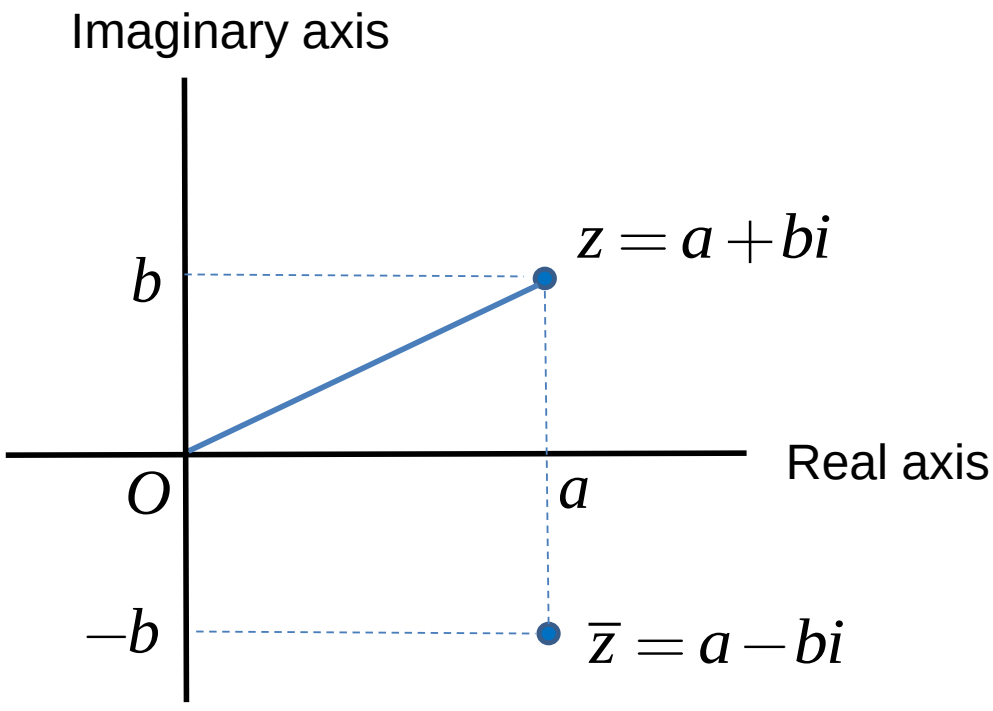
The *absolute value* (or *modulus*) of z is

$$|z| = \sqrt{z\bar{z}} = \sqrt{a^2 + b^2}.$$

We note that if $z = a$ is a real number, then $|z| = \sqrt{a^2} = |a|$, which is the ordinary absolute value of a real number, and is equal to the distance between the point a and the origin on the real line.
Since a complex number $z = a + bi$ corresponds to a pair (a, b) of real numbers, it can be represented by a point in the plane.

Appendix B.5 Complex numbers

Geometric interpretation The distance between the point (a, b) and the origin is the absolute value (modulus) of $a + bi$.



$\overline{z}$ is the mirror image of z about the real axis,

$|z|$ is the distance between the point z and the origin.

Appendix B.5 Complex numbers

Example $(1+i)(1-2i) = 1-2i+i-2i^2 = 3-i.$
Example Let $z = 2-i$. Then

$$\text{Re } z = 2,\ \text{Im } z = -1,\ \bar{z} = 2+i,$$
$$|z| = \sqrt{2^2+(-1)^2} = \sqrt{5}.$$

We may divide a complex number by another nonzero complex number and express the quotient in standard form.
Example Express the quotient $\frac{i}{1-i}$ in standard form.
Solution We multiply the denominator and the numerator by the conjugate of the denominator:

$$\frac{i(1+i)}{(1-i)(1+i)} = \frac{i+i^2}{2} = \frac{-1}{2} + \frac{1}{2}\,i.$$

Appendix B. Complex numbers

The following are some useful properties about the conjugate and absolute values of complex numbers.

i) $\bar{z} = z$ if and only if z is a real number.

ii) $\overline{w + z} = \bar{w} + \bar{z}$.

iii) $\overline{wz} = \bar{w} \cdot \bar{z}$.

iv) $z\bar{z} = |z|^2 \geq 0$, $|z| = 0$ if and only if $z = 0$.

v) $|wz| = |w||z|$.

vi) $\left|\dfrac{w}{z}\right| = \dfrac{|w|}{|z|}$.

(vii) $\overline{\left(\dfrac{w}{z}\right)} = \dfrac{\bar{w}}{\bar{z}}$.

Appendix B.5 Complex numbers

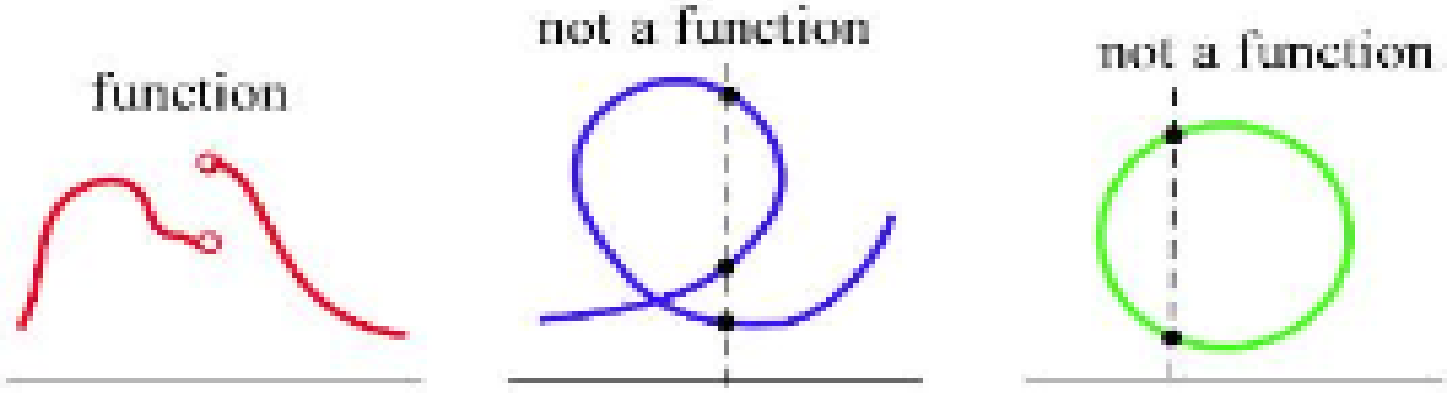
Example Let $w = 1 + 2i$ and $z = 2 - i$. Then

$$\begin{aligned}
\bar{w} &= 1 - 2i, \ \bar{z} = 2 + i, \\
|w| &= \sqrt{1^2 + 2^2} = \sqrt{5}, \ |z| = \sqrt{2^2 + (-1)^2} = \sqrt{5}, \\
wz &= 2 + 4i - i - 2i^2 = 4 + 3i, \\
\frac{w}{z} &= \frac{1 + 2i}{2 - i} = \frac{(1 + 2i)(2 + i)}{(2 - i)(2 + i)} = \frac{(1 + 2i)(2 + i)}{5} \\
&= \frac{(2 - 2) + (1 + 4)i}{5} = i, \\
|wz| &= \sqrt{4^2 + 3^2} = 5, \\
\left|\frac{w}{z}\right| &= \sqrt{0^2 + 1^2} = 1, \\
\frac{\bar{w}}{\bar{z}} &= \frac{1 - 2i}{2 + i} = \frac{(1 - 2i)(2 - i)}{(2 + i)(2 - i)} = \frac{-5i}{5} = -i.
\end{aligned}$$

The Vertical Line Test

The Vertical Line Test A curve in the xy-plane is the graph of a function if and only if each vertical line intersects the curve in at most one point.

Example Which of the following curves are graphs of functions?



function not a function not a function

2.2 Algebra of Functions

New functions can be formed use *addition*, *subtraction*, *multiplication*, *division* of known functions in an obvious way:

$$\begin{aligned}(f+g)(x) &= f(x)+g(x),\\ (f-g)(x) &= f(x)-g(x),\\ (fg)(x) &= f(x)g(x),\\ (f/g)(x) &= f(x)/g(x).\end{aligned}$$